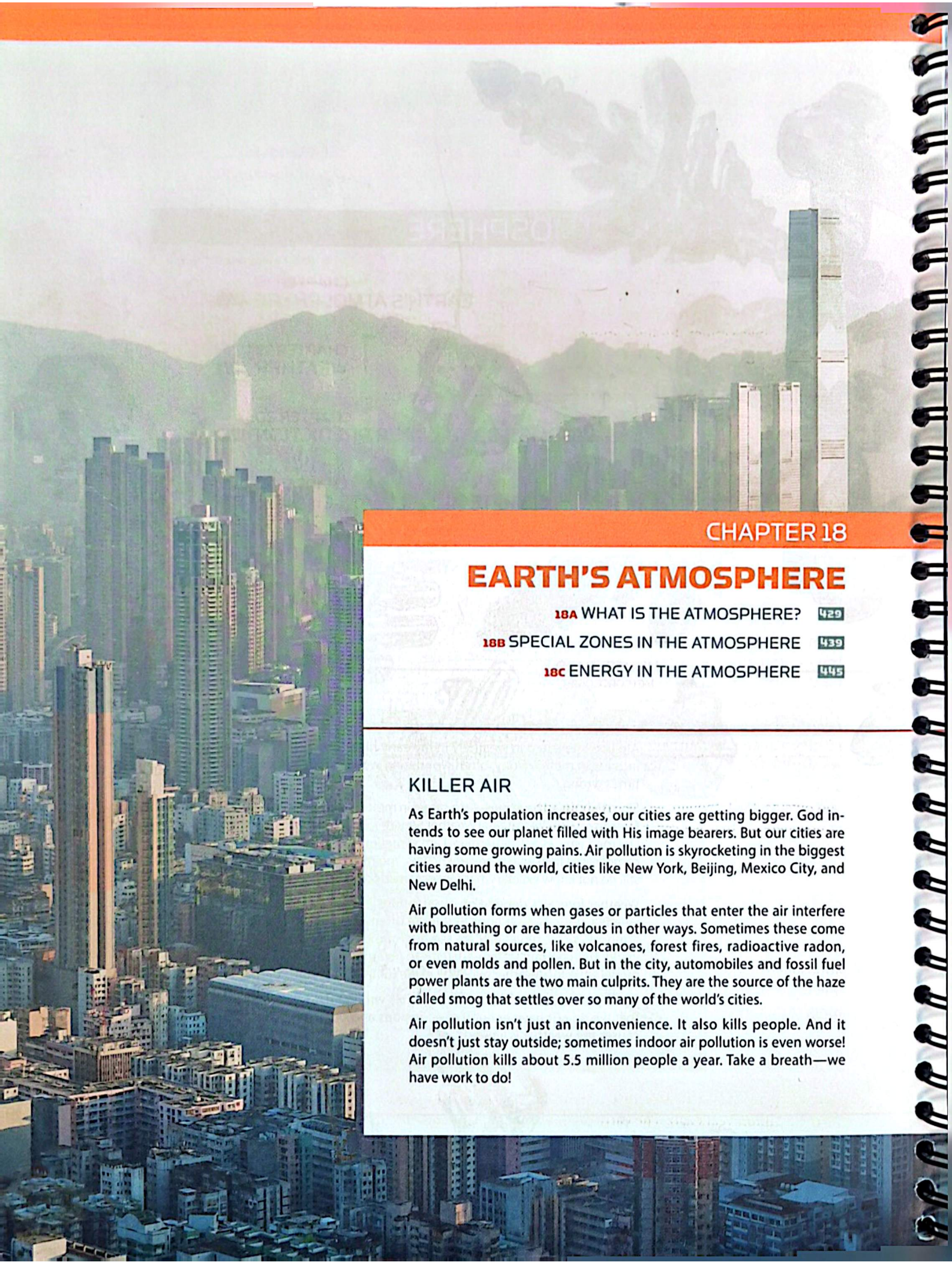


CHAPTER 18

EARTH'S ATMOSPHERE

18A WHAT IS THE ATMOSPHERE? 429

18B SPECIAL ZONES IN THE ATMOSPHERE 439

18C ENERGY IN THE ATMOSPHERE 445

KILLER AIR

As Earth's population increases, our cities are getting bigger. God intends to see our planet filled with His image bearers. But our cities are having some growing pains. Air pollution is skyrocketing in the biggest cities around the world, cities like New York, Beijing, Mexico City, and New Delhi.

Air pollution forms when gases or particles that enter the air interfere with breathing or are hazardous in other ways. Sometimes these come from natural sources, like volcanoes, forest fires, radioactive radon, or even molds and pollen. But in the city, automobiles and fossil fuel power plants are the two main culprits. They are the source of the haze called smog that settles over so many of the world's cities.

Air pollution isn't just an inconvenience. It also kills people. And it doesn't just stay outside; sometimes indoor air pollution is even worse! Air pollution kills about 5.5 million people a year. Take a breath—we have work to do!

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18A

WHAT IS THE ATMOSPHERE?

How does the atmosphere change with elevation?

18A Objectives

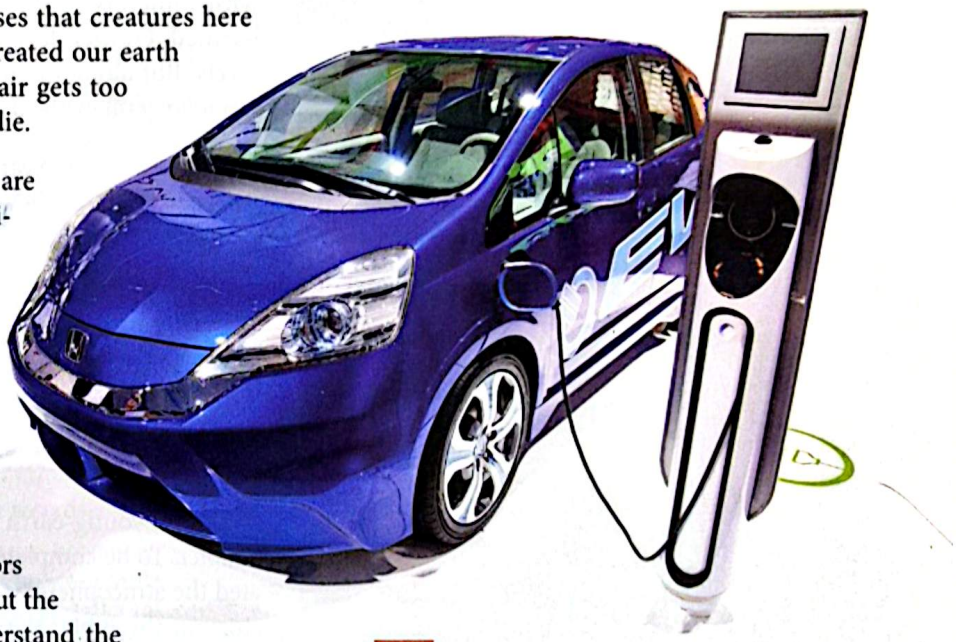
After completing this section, you will be able to

- describe how people can affect the atmosphere.
- identify evidence of design in the atmosphere.
- sketch the atmosphere's composition, temperature, and structure.
- trace the flow of carbon and nitrogen in the atmosphere.

18.1 AIR CARE

Our **atmosphere**, the envelope of gases that surrounds our planet, is crucial to life on Earth. Consider some of the other planetary bodies in our solar system. Jupiter's atmosphere is mostly hydrogen and helium, with clouds of ammonia ice crystals. Have fun trying to get a breath of fresh air there. You're not going to do much better on Mars or Venus, which have atmospheres made mostly of carbon dioxide. Venus's so-called air contains sulfuric acid! Closer to home, the moon is even worse. It hardly has a measurable atmosphere at all. Earth's atmosphere provides the gases that creatures here need to live. The Bible says that God created our earth to be inhabited (Isa. 45:18). But if our air gets too polluted, living things can get sick and die.

We need to take care of our air. Scientists are looking for ways to reduce the toxic particles and harmful gases that people release into the air. Cutting back on *emissions* from manufacturing, from cars and trucks, and from coal-fired power plants is part of this. Developing vehicles and power plants that don't use fossil fuels at all is another way, though some pollutants are unavoidable. But we should try to work on ways to filter and clean the air we breathe, both indoors and outside. First, though, let's learn about the atmosphere so that we can better understand the challenges we face.



18-1
Developing technology like this electric car can help reduce air pollution in cities around the world.

18.2 ORIGIN OF THE ATMOSPHERE

Some people credit the marvelous uniqueness and design of Earth's atmosphere to pure chance. To them, our amazing atmosphere is just one big coincidence. Do you remember their view of Earth's beginnings from Chapter 5?

THE OLD-EARTH STORY

According to old-earth scientists, the earth began forming more than 4.5 billion years ago. The early earth was a smaller, molten ball covered by a thin crust. Within several hundred million years of the planet's birth, a moon-forming collision occurred and melted the

Jupiter : mostly hydrogen and helium, with clouds of ammonia ice crystals.

Mars or Venus : mostly made of CO_2

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entire earth again. As the earth cooled from this collision, gaseous materials gathered into an atmosphere that was hot and contained mainly the light gases hydrogen and helium, which on their own can't support life. No free oxygen was present in the early atmosphere. The emissions of numerous volcanoes changed its composition. Water vapor together with carbon and nitrogen compounds began to collect. Ice from comet bombardments added water and other compounds to the oceans and atmosphere.

Sometime during the first billion years of Earth's existence, evolutionists believe that life appeared from nonlife. These life forms were *anaerobic bacteria* that could live in the absence of oxygen. Biochemists believe that oxygen would have destroyed the complex molecules needed to build even the simplest cell. Even so, ultraviolet light from the sun broke down some water vapor into oxygen and hydrogen. This oxygen quickly reacted with minerals in the rocks, so the simple bacteria were not harmed. Oxygen levels were probably less than 1% of today's levels. But along the way, oxygen-producing organisms called *cyanobacteria* evolved.

Around 2.4 billion years ago, the surface rocks had reacted with as much oxygen as they could, so the extra oxygen collected in the atmosphere, and the oxygen levels in the air jumped. This *oxygen catastrophe* killed off most of the anaerobic organisms, and paved the way for more advanced forms of life to evolve. Secular scientists believe that photosynthetic organisms are responsible for the modern oxygen-rich atmosphere we have today.

THE YOUNG-EARTH STORY

A biblical young-earth view of the atmosphere's origin is much simpler. To be completely honest, we don't *know* when God created the atmosphere because it is not mentioned in the Creation story in the Bible. We can assume that when God formed the firmament (or great expanse) between the waters above and the waters below on Day 2 (Gen. 1:6–8), He also created the atmosphere in preparation for the creation of plants on Day 3. Plants needed atmospheric carbon dioxide and water vapor to live, and all the other creatures and humans definitely needed to breathe during the remaining three days of the Creation week.

God made the atmosphere "very good," along with the rest of His creation. Even the Fall probably didn't noticeably change the atmosphere. But the Flood did. Volcanic activity and changes in the sizes and shapes of the oceans deeply affected the earth's atmosphere. We can still see the effects of these changes today in the atmospheric chemistry, weather patterns, and climate zones all over the world. But though God judged our world, it still bears signs of His design and loving care. The atmosphere is one grand exhibit of this.

BI Experiment in Evolution

Have you heard of the Urey-Miller experiment? Two evolutionists did an experiment in which they simulated how life could have evolved in Earth's early atmosphere. They took a mixture of hydrogen, carbon dioxide, methane, water vapor, and ammonia and sparked electricity in it. They produced what they called "the precursors of life." Evolutionists called it a victory for their theory. But there are some major problems with this experiment.

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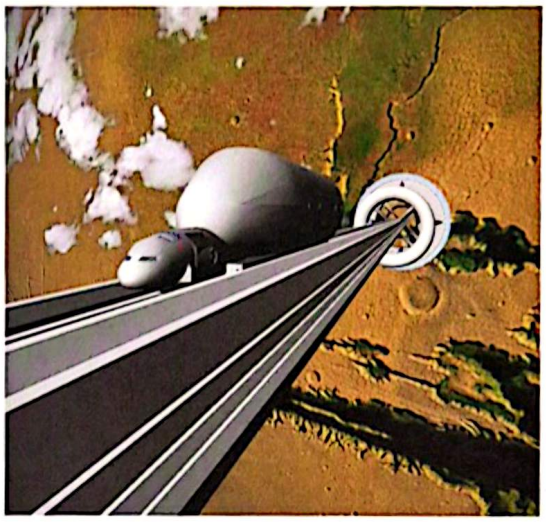
18.3 ATMOSPHERIC REGIONS BY COMPOSITION

You are a pioneering meteorologist, taking a trip to explore Earth's atmosphere from the ground to space in person! You are one of the first passengers in NASA's new *space elevator*. But you need some information about the atmosphere before your trip. Will you need supplemental oxygen? How should you dress? Sounds like you will learn much about the layers of the atmosphere!

We can study the earth's atmosphere in several different ways. For example, we may be interested in its *composition*, that is, the kinds of gases it contains. In Chapter 2, you learned how everything is made of atoms. Different atoms make up different chemical elements. Nitrogen makes up more than three-fourths of the atmosphere near the earth's surface. It doesn't easily react with other chemicals, so it exists as a pure element in air and isn't poisonous. One of its main purposes is to dilute oxygen to the concentration that is best for life while avoiding a fire hazard! This is not an accident of Earth's geologic history; it is evidence of God's wonderful design.

18-2

First proposed in 1895, a space elevator would allow movement from Earth's surface to space without the use of rockets. This is an artist's rendering of an elevator car traveling to space.



CASE STUDY: SKYDIVING FROM SPACE

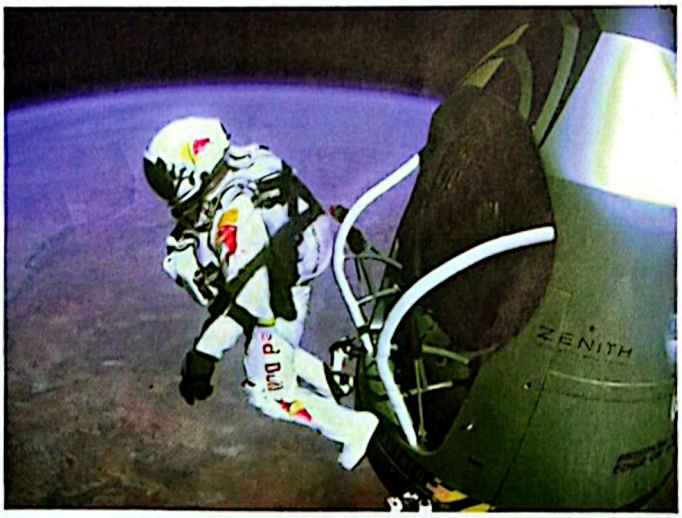
Looking down toward Earth from your space elevator, you might be tempted to jump if you had a parachute. Joseph Kittinger actually jumped from the upper atmosphere in 1960! Riding a balloon to 31,300 m, he skydived from this height. He set the record for the highest and fastest parachute jump in history as well as the longest free fall. It was fifty-two years before anyone even attempted to break these records.

On October 14, 2012, Felix Baumgartner began his day by donning a space suit. He rose to 39,040 m in a space capsule held aloft by a balloon. He jumped from this height to set the record for the highest skydive. Plummeting 36,529 m in 4 minutes and 20 seconds, he also set a new record for the longest free fall. He set another record for being the fastest skydiver at 1345 kph. Baumgartner broke the sound barrier during his free fall, exactly sixty-five years after Chuck Yeager accomplished the same feat in the Bell X-1 aircraft. However, Baumgartner failed to break the record for the longest time of descent. Joseph Kittinger participated in Baumgartner's record-breaking event as a member of mission control.

Both of these jumps show how far (or high) man will go to study the world God gave us. Data from each jump provided information about both the atmosphere and how people handle stresses in this environment. Engineers can use this information to design systems to protect people from these harsh surroundings.

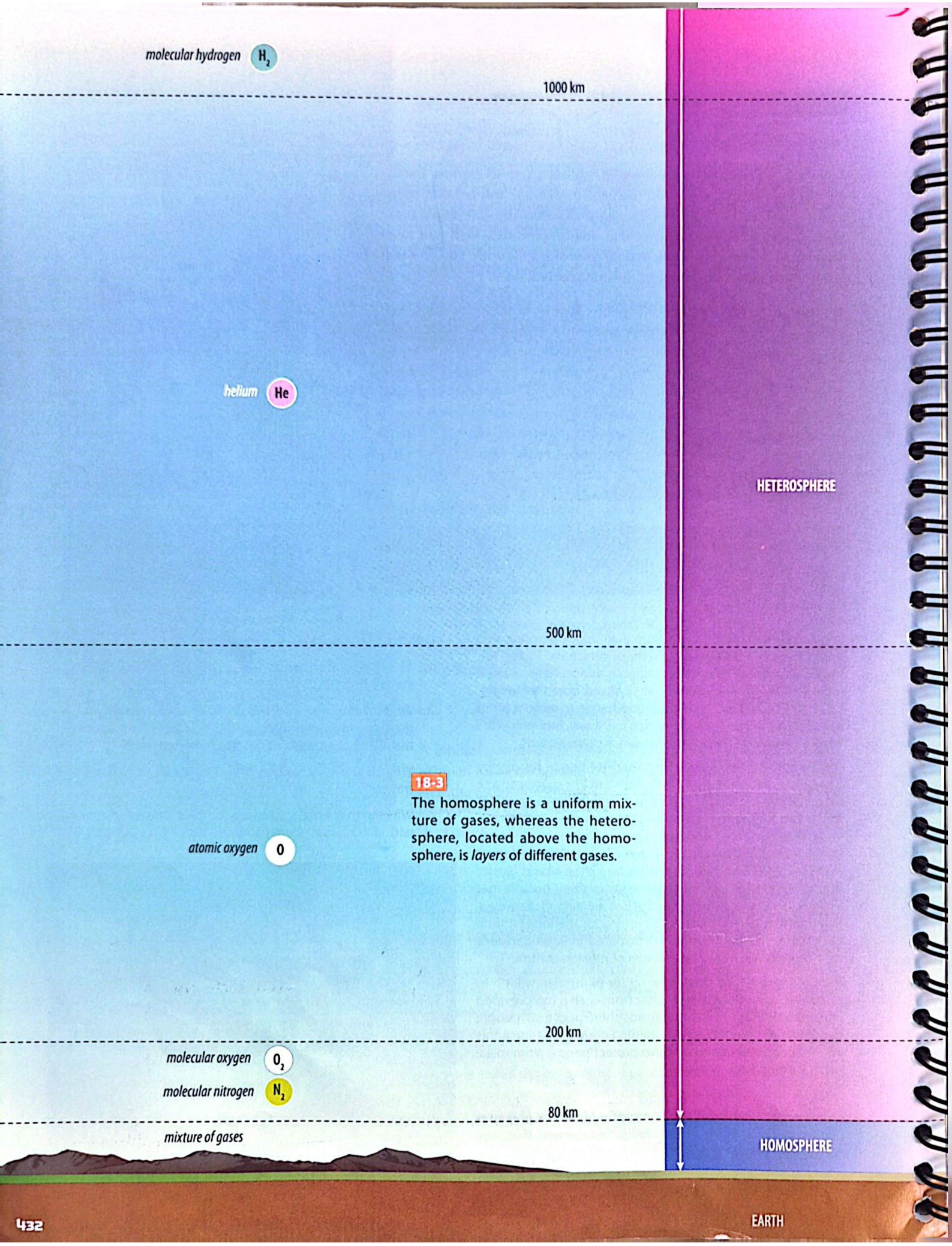
Questions to Consider

1. At 39,040 m, from which layer of the atmosphere did Felix start his jump?
2. Where within the layers of the atmosphere would these men have traveled through the coldest air? At approximately what altitude would this have occurred?
3. What is the composition of the air through which they traveled?
4. Why did they need space suits for their jumps if they were within the atmosphere the entire time?



Felix Baumgartner leaves the capsule for his record-breaking skydive in 2012. Notice the protective equipment that was needed even though they were still in Earth's atmosphere.

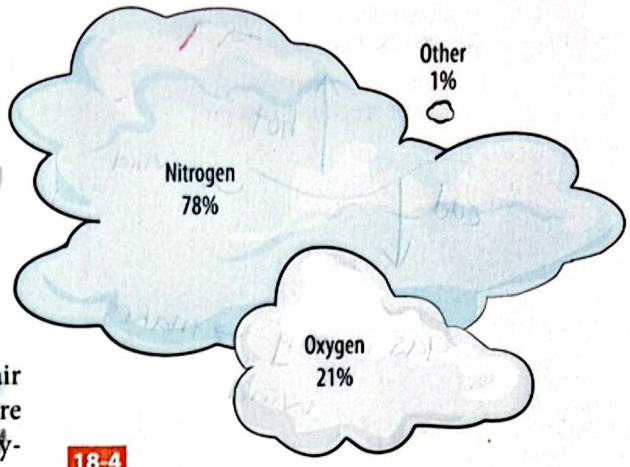
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Carbon dioxide ⇒ between nearly 0 %. to a little more than 2 %.
Water vapor ⇒ almost 0 %. to 5 %.

We know that the atmosphere near the ground is a mixture of gases. Its volume is about 21% oxygen, 78% nitrogen, and 1% other gases such as argon, carbon dioxide, helium, and hydrogen. The portions of all these gases near the earth's surface are almost constant, except for carbon dioxide and water vapor. Carbon dioxide may vary between nearly 0% to a little more than 2%. Water vapor may vary from almost 0% to 5%.



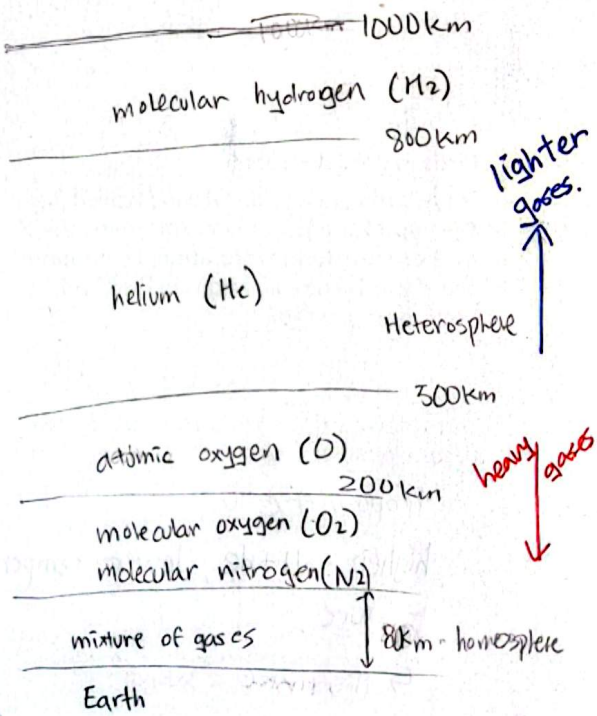
18-4 Air is mostly nitrogen, which serves to dilute the oxygen.

As we travel upward in our space elevator we notice that the composition of the atmosphere changes. All the gases that make up air are thoroughly mixed together in the lowest part of the atmosphere by winds and weather. They don't separate, settle out, or form layers. We call this uniform mix of gases the **homosphere**. It stretches upward to an altitude of 80 km. However, starting at about 3 km, you would need to breathe pressurized oxygen. Above 3 km, there is too little pressure for air to supply all the oxygen you need. Your brain starts acting fuzzy and your reactions slow down. That's why you are wearing a space suit for your trip!

The prefix *homo-* means "like," "similar," or "uniform." The **homosphere** layer of the atmosphere is a nearly uniform mixture of gases.

As we travel above the homosphere, air composition changes. In this region, called the **heterosphere**, the atmosphere has several layers of different gases. Since there is little wind at these altitudes, gases settle out by their atomic weights—with the heaviest gases nearer the bottom. From lowest to highest altitude, they are molecular nitrogen (N_2), molecular oxygen (O_2), atomic oxygen (O), helium (He), and molecular hydrogen (H_2). You can see the arrangement of these layers in Figure 18-3 on the facing page. This order makes sense, since the nitrogen molecule is fourteen times as heavy as the hydrogen molecule.

The prefix *hetero-* means "separate" or "different." The **heterosphere** layers of the atmosphere contains separate layers of distinct gases.



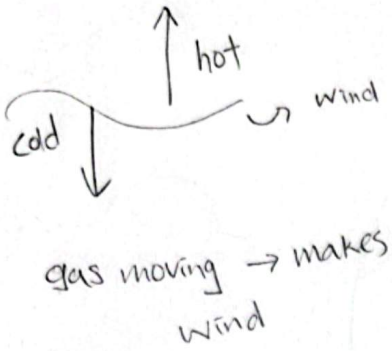
18.4 TEMPERATURE LAYERS OF THE ATMOSPHERE

In addition to changes in composition, you also notice that the temperature probe on the exterior of the space elevator is indicating significant change as you travel upward through the atmosphere. The atmosphere has five distinct temperature layers. Air temperature changes within a layer, warming or cooling with altitude at different rates.

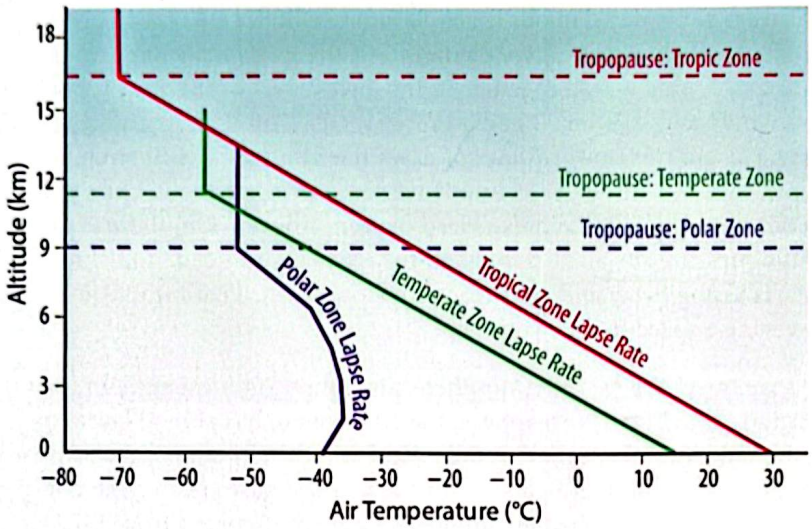
The lowest layer of the atmosphere is the **troposphere** (TROE puh SFEER). This zone begins at the earth's surface and extends upward to about 9 km over the North and South Poles, to about 11–12 km over the continental United States, and to 16–17 km over the Equator. This is where most of Earth's weather happens. As your elevator climbs, you will pass through layers of clouds. You can expect to run into brief rain showers. Gusts of wind will swing your space elevator back and forth on its carbon tether. In the troposphere, temperature drops with increasing altitude. The temperature lowers about 6.5°C for every kilometer increase in altitude. Atmospheric scientists call this kind of steady change in temperature a **lapse rate**. At approximately 11 km up, the temperature stops changing. This upper boundary of the troposphere is the *tropopause*. Here, the temperature is a chilly –55°C. Brrr! Your elevator's electrical heating and thermal insulation will pay off.

tropopause: tropo- (to turn or change) + -pause (L. *pausa*—stop)

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The cold air at the top of the troposphere is dense and tends to sink. The warm air at the bottom of the troposphere is less dense, rises, and is replaced by the colder air settling from above. This process constantly mixes air in the troposphere. Rising warm air carries moisture that forms clouds and rain. Sinking cold air brings clear, cooler weather. This kind of thermal mixing happens only in the troposphere.



18-5
The steady drop in temperature as altitude increases is called the *lapse rate*. For regions of the atmosphere where the temperature *rises*, the lapse rate is *negative*.

stratosphere: (STRAT uh SFEER): strato- (L. *stratus*—a spreading out or layer) + -sphere

But your elevator is still climbing! You notice that the stratosphere, the second temperature layer, extends from the top of the troposphere to about 50 km. The stratosphere is free of clouds and dust. In this layer, the temperature *increases* from about −55 °C at the bottom, to about 0 °C at the top. You'll be able to see for hundreds of kilometers from this altitude. Aircraft flying long distances share the stratosphere with your vehicle to avoid turbulent storms in the troposphere. It can be windy here too. Some winds in the stratosphere blow as fast as 400 kph in paths called **jet streams**. Aircraft sometimes fly with these jet streams to increase their speed and to save fuel. Jet streams sometimes temporarily dip into the troposphere. The upper boundary of the stratosphere is the *stratopause*.

Birds in the Jet Stream?
Some birds catch a ride on winds called *thermals* up to the jet stream. This helps them save energy during migration. Learn more about winds and migrations in the Life Connection on page 463.

mesosphere (MES uh SFEER): meso- (Gk. *mesos*—middle) + -sphere

As the elevator travels into the mesosphere, you notice that air temperature again decreases with altitude. The coldest temperature of the atmosphere, −90 °C, occurs at the top of the mesosphere. Like the troposphere and the stratosphere, the mesosphere is a uniform mix of gases. Also like the lower layers, the mesosphere has winds. These winds often have high speeds and come from different directions. The mesosphere ends at the *mesopause* about 80 km above the earth. The troposphere, stratosphere, and mesosphere all lie within the homosphere composition layer (see Figure 18-3).

troposphere
higher altitude, lower temperature.
↳ this make "wind"

discovery
mesosphere → no moisture so no weather.
troposphere → has moisture so there is weather.

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How does stratosphere? ↑ wind exsits

↳ There are only warm air so it stays up. but earth is moving, so it makes wind.

temperature profile during minimal sunspot activity

temperature profile during maximum sunspot activity

EXOSPHERE

500 km

International Space Station

18-6 Atmospheric layers classified by temperature

atomic oxygen auroras

THERMOSPHERE

molecular oxygen auroras

meteors

molecular nitrogen auroras

80 km

MESOSPHERE

Homosphere.

50 km

STRATOSPHERE

fighter jet

11 km

Mount Everest

TROPOSPHERE

−200 0 200 400 600 800 1000 1200

Temperature (°C)

435

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18-7
Noctilucent clouds, strange bluish-white clouds, appear in the mesosphere. Meteorologists aren't sure where they come from, but they're much different from clouds in the troposphere.

thermosphere (THURM uh SFEER): thermo- (Gk. *thermos*—warm) + -sphere

Passing through 80 km, your elevator enters the last true layer of the atmosphere—the **thermosphere**, which extends to the *thermopause* at about 500 km. The atmosphere is very thin here. Just 20 km above the mesopause, you enter true space at about 100 km. Higher than this altitude, an aircraft would have to move faster than orbital speed around the earth to get usable lift from the air. Scattered atmospheric atoms and molecules absorb the sun's full radiation. Daytime temperatures range from 650 °C to 1250 °C at the highest part of the thermosphere. In spite of the high temperatures, you would not feel warm in the thermosphere as long as you stayed in the shadow of your elevator vehicle. The molecules are so widely separated that too few would contact your suit to transfer heat that you could feel. So warmth from the surrounding daytime atmosphere is negligible. However, you would cook in direct sunlight without an insulated space suit!

Spacecraft Cooling

Spacecraft traveling through the thermosphere during the day get hot from absorbing solar radiation. Because objects at this altitude must move at high speeds to stay in orbit, friction with the thin air makes them heat up even more.

To avoid absorbing excess heat, spacecraft are painted white or covered in mirrored skins. Both methods reflect heat away from the vehicle. Some spacecraft have fins that radiate heat into space on the shadowed side of the vehicle. Many satellites rotate to spread heating over their entire surfaces.

Figure 18-6 shows daytime temperatures for the thermosphere. At night, the temperatures can be hundreds of degrees lower. Though the gas particles making up the thermosphere are widely scattered, they block some harmful radiation. They also protect the earth from small space rocks called *meteoroids* by burning them up as "shooting stars."

So what's above the thermosphere? There is no clear line between Earth's atmosphere and interplanetary space. Instead, the atmosphere becomes thinner as the distance from the earth increases until it has the same number of molecules per cubic meter as space itself. The transitional zone—not a true layer—between the atmosphere and interplanetary space is the **exosphere**. Gases in this layer can escape from Earth's gravity into space, which begins near the bottom of the atomic oxygen layer of the atmosphere. Gas particles are so far apart here it's difficult to measure their temperature.

exosphere (EX uh SFEER): ex- (Gk. *exo*—outside) + -sphere

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18.5 CYCLES AND THE ATMOSPHERE

As you have learned in earlier chapters, key chemicals needed by living things are recycled in the natural environment. For example, organisms require carbon, nitrogen, and water, among other things. If these resources were not replenished by recycling, supporting the present amount of life on Earth would not be possible. In Chapter 13, we discussed how these nutrients flow from place to place in the marine environment. But that's just part of the big picture. What role does the atmosphere play in global nutrient cycles?

THE CARBON CYCLE

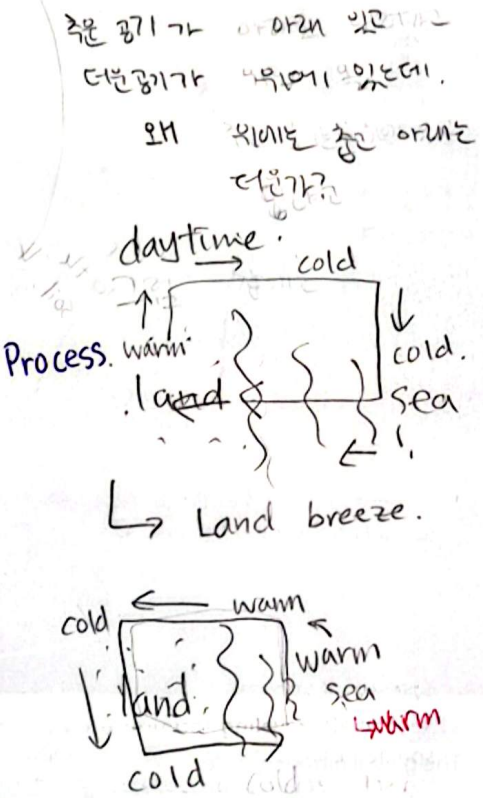
The flow of carbon from living things to the carbon reservoirs in the environment and back again is what the *carbon cycle* is all about. Atmospheric carbon dioxide is a key carbon reservoir in the carbon cycle. Plants use the gas during photosynthesis to build their tissues and produce fruits made of complex carbon compounds. When animals and people eat the plants, they take in the carbon compounds, which become part of *their* tissues. Some of the food we eat is broken down for energy, and, as part of this process, carbon dioxide is exhaled into the atmosphere.

After living things die and decay, their carbon compounds enter the soil carbon reservoir. Remember that carbon dioxide and carbon compounds can dissolve in surface water and groundwater, and can ultimately enter lakes and even the ocean. The freshwater and marine carbon cycles both take part in the overall movement of carbon through the environment. So there is a flow of carbon between living things, the earth, water, and the atmosphere.

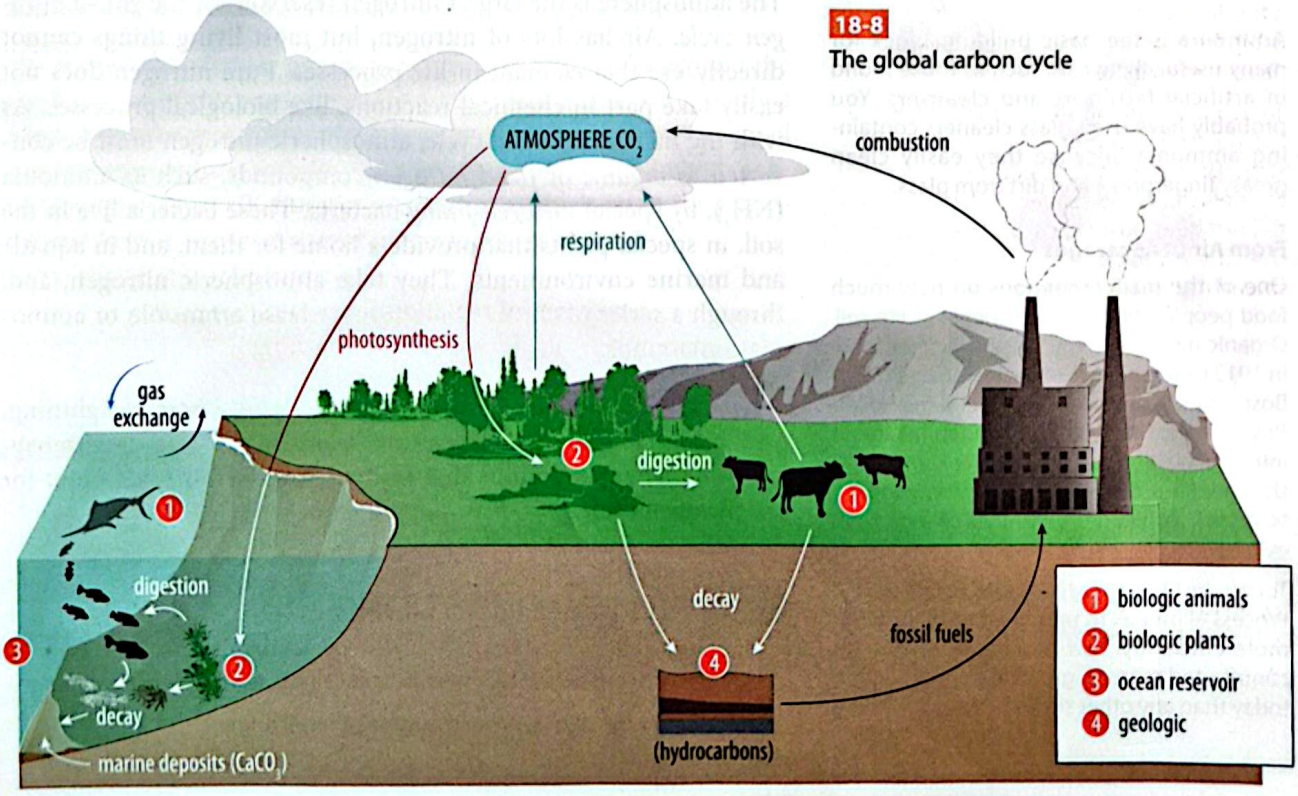
Q : Cold air ↓ Warm air ↑

Why does air ↓

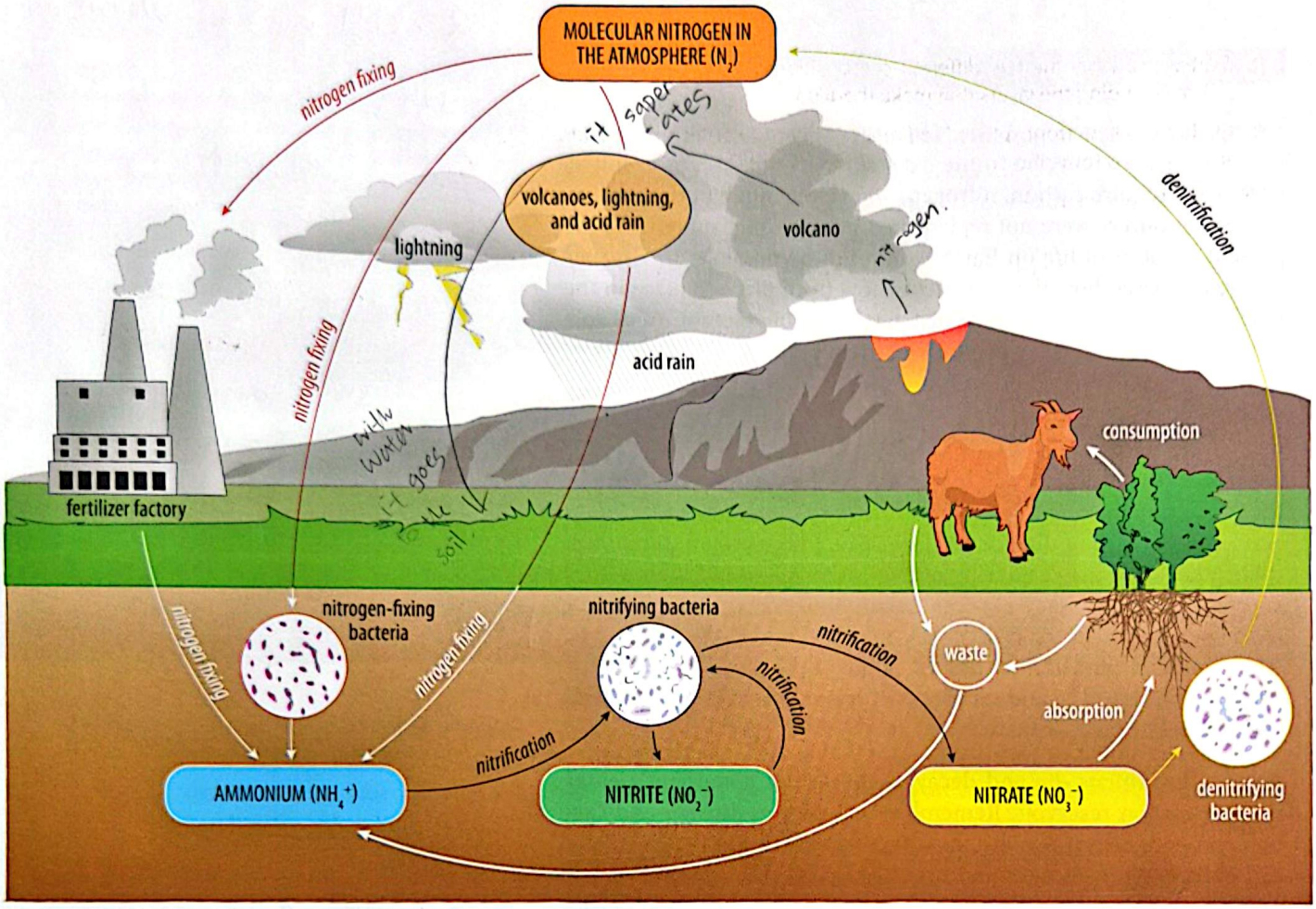
latitude ↑



18-8
The global carbon cycle



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18-9
The global nitrogen cycle

Ammonia is the basic building block for many useful chemicals, such as those found in artificial fertilizers and cleansers. You probably have used glass cleaners containing ammonia because they easily clean greasy fingerprints and dirt from glass.

From Air to Asparagus

One of the main limitations on how much food people can grow is nutrients in the soil. Organic nitrogen is key, since it is plant food. In 1913 German chemists Fritz Haber and Carl Bosch introduced a method, called the *Haber-Bosch Process*, that turns nitrogen in the air into artificial fertilizer. This is the same thing that bacteria do! Suddenly, it was possible to grow much larger quantities of food than ever before.

Today, fertilizer made by the Haber-Bosch Process allows us to produce food four times more efficiently. This process probably has contributed more to growing the population today than any other single human discovery.

THE NITROGEN CYCLE

The atmosphere is the largest nitrogen reservoir for the global *nitrogen cycle*. Air has lots of nitrogen, but most living things cannot directly use that element in life processes. Pure nitrogen does not easily take part in chemical reactions, like biological processes. As with the marine nitrogen cycle, atmospheric nitrogen must be converted to *organic* or *fixed nitrogen* compounds, such as ammonia (NH_3), by special *nitrogen-fixing* bacteria. These bacteria live in the soil, in special plants that provide a home for them, and in aquatic and marine environments. They take atmospheric nitrogen, and, through a series of chemical reactions, release **ammonia** or ammonia compounds.

Another source of fixed nitrogen in the atmosphere is lightning. Lighting breaks down nitrogen in the air and combines it with oxygen. The new compounds that form are another starting point for the nitrogen cycle.

18A REVIEW QUESTIONS

1. What are some ways that we can reduce pollutants in the atmosphere?
2. Compare the two views of the atmosphere's origin.

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3. Sketch and label the two different composition regions of the atmosphere, including the layers that make them up. 432pg. 그림.

4. Explain how the troposphere is different from all the other temperature layers in the atmosphere. weather happens.

5. How can we exercise wise dominion by using the jet streams in the stratosphere? tro

6. How is the mesosphere unique among the temperature layers of the atmosphere? mesosphere has coldest temperature.

7. Which true layer of the atmosphere has the hottest daytime temperatures? thermosphere

8. Why can't plants get nitrogen directly out of the air to use for building tissues? → atmospheric nitrogen inerts.. ↳ in active.

9. (True or False) Atmospheric temperature doesn't decreases continuously and smoothly with altitude from the ground to the exosphere.

18B

SPECIAL ZONES IN THE ATMOSPHERE

How do special layers of the atmosphere protect life on Earth?

There are a few zones in the atmosphere that are pretty special. They do some very important things for the earth and help improve our ability to put dominion into practice. Their presence suggests that God designed them as features of a very good earth. Let's take a look at them.

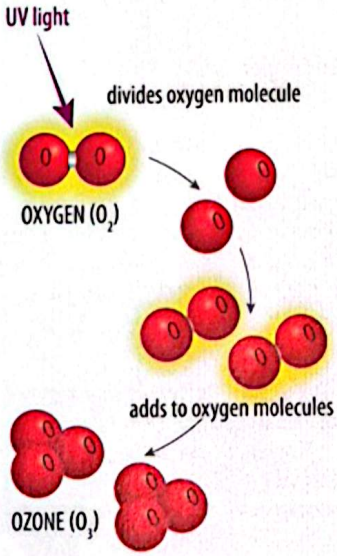
18B Objectives

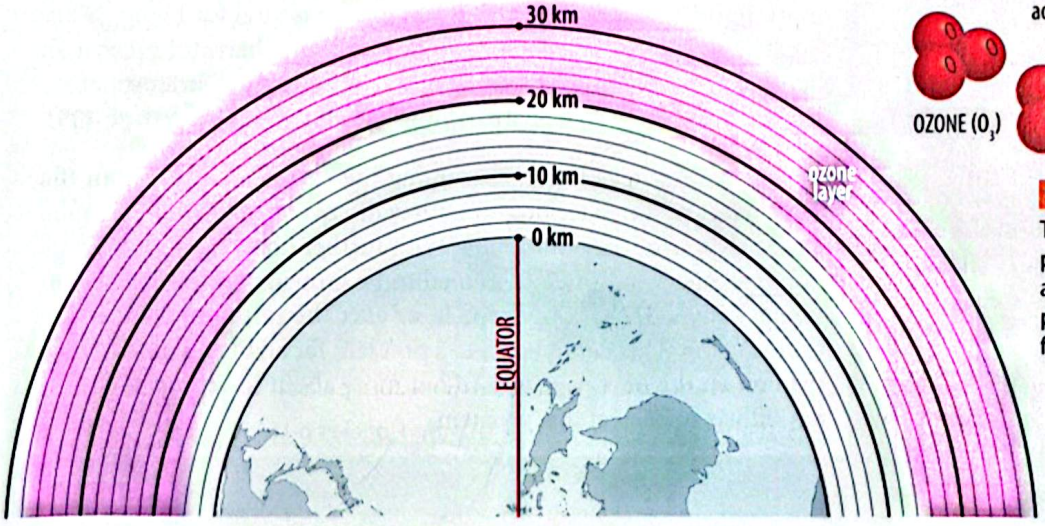
After completing this section, you will be able to

- relate special zones of the atmosphere to the other layers.
- explain how the special zones in the atmosphere are evidence of God's good design.

18.6 OZONE LAYER

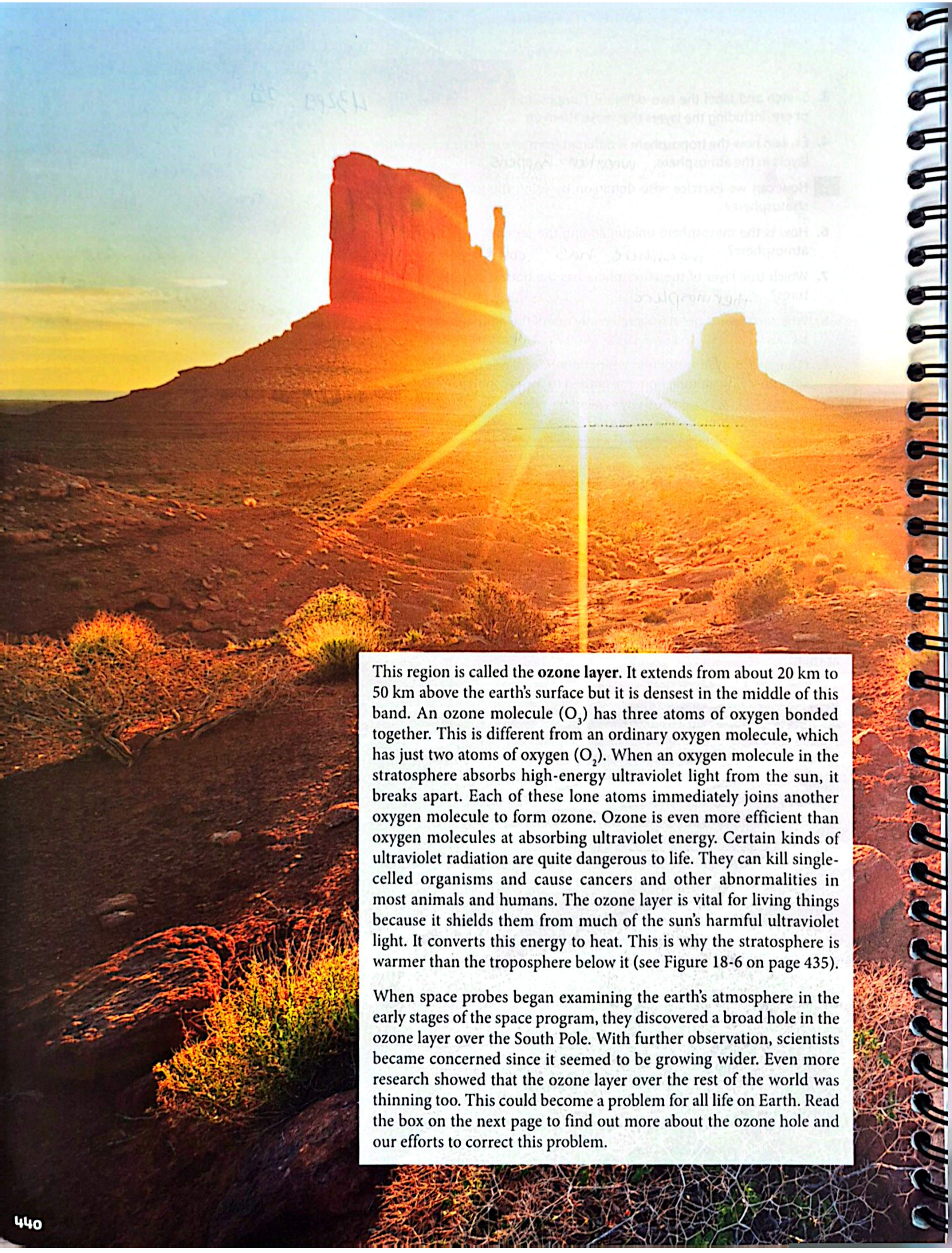
We started this chapter mentioning smog in large cities around the world. A lot of this smog contains a gas called *ozone*. It's actually a kind of oxygen, but you couldn't breathe it for long. Ozone in small amounts causes lung irritation and breathing problems. In high concentrations, it can kill. But did you know that high up in the stratosphere there is a layer of ozone? There it actually helps life!





18-10
The altitude of the densest part of the ozone layer in the atmosphere. Ozone's unique properties shield the earth from the sun's ultraviolet light.

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This region is called the **ozone layer**. It extends from about 20 km to 50 km above the earth's surface but it is densest in the middle of this band. An ozone molecule (O_3) has three atoms of oxygen bonded together. This is different from an ordinary oxygen molecule, which has just two atoms of oxygen (O_2). When an oxygen molecule in the stratosphere absorbs high-energy ultraviolet light from the sun, it breaks apart. Each of these lone atoms immediately joins another oxygen molecule to form ozone. Ozone is even more efficient than oxygen molecules at absorbing ultraviolet energy. Certain kinds of ultraviolet radiation are quite dangerous to life. They can kill single-celled organisms and cause cancers and other abnormalities in most animals and humans. The ozone layer is vital for living things because it shields them from much of the sun's harmful ultraviolet light. It converts this energy to heat. This is why the stratosphere is warmer than the troposphere below it (see Figure 18-6 on page 435).

When space probes began examining the earth's atmosphere in the early stages of the space program, they discovered a broad hole in the ozone layer over the South Pole. With further observation, scientists became concerned since it seemed to be growing wider. Even more research showed that the ozone layer over the rest of the world was thinning too. This could become a problem for all life on Earth. Read the box on the next page to find out more about the ozone hole and our efforts to correct this problem.

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THE OZONE HOLE

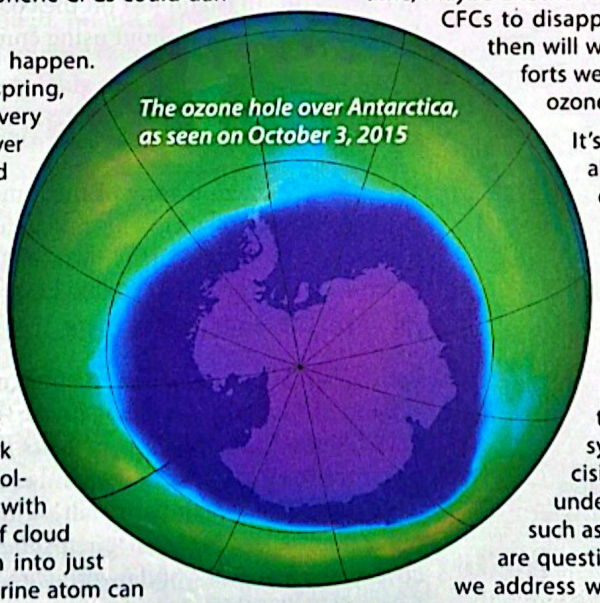
Ozone is the poisonous gas that we can't live without. It forms a layer high in the stratosphere called the *ozone layer*. This layer actually protects life on Earth by absorbing harmful ultraviolet rays from the sun.

However, there is a problem. Every spring, the ozone layer develops a hole above Antarctica. It's a good thing there aren't too many people there! Scientists have linked this hole to a class of chemicals.

The main culprit is dichlorodifluoromethane (CCl_2F_2), otherwise known as Freon. It was developed in 1930 by Thomas Midgley Jr. and Charles Kettering of General Motors. People used it in refrigerators, air conditioners, dry cleaning solvents, aerosols, and some kinds of foam. Freon and other related chemicals called chlorofluorocarbons (CFCs) are odorless, tasteless, nonflammable, and noncorrosive. They are also not poisonous in small amounts.

In the 1950s, scientists studying the atmosphere discovered the ozone layer and its shielding effect. In the early 1970s, chemists found that ozone breaks down in the presence of CFCs. They predicted that atmospheric CFCs could dangerously deplete the ozone layer.

It takes several things for this to happen. First, you need clouds. In the spring, clouds of ice crystals form in the very high cold air of the stratosphere over Antarctica. These ice crystals speed up certain chemical reactions. To deplete the ozone, you also need CFCs. These chemicals contain chlorine, along with other elements. Sunlight is the third ingredient. If you remember, Antarctic winters see no sun. Not until springtime does the sun reach these areas of the world. When it does, energy in sunlight can break off a chlorine atom from a CFC molecule. This chlorine atom reacts with ozone molecules on the surface of cloud ice crystals to break them down into just plain old oxygen. One single chlorine atom can react over and over again. So even a little CFC can do a lot of damage.



The ozone hole over Antarctica, as seen on October 3, 2015

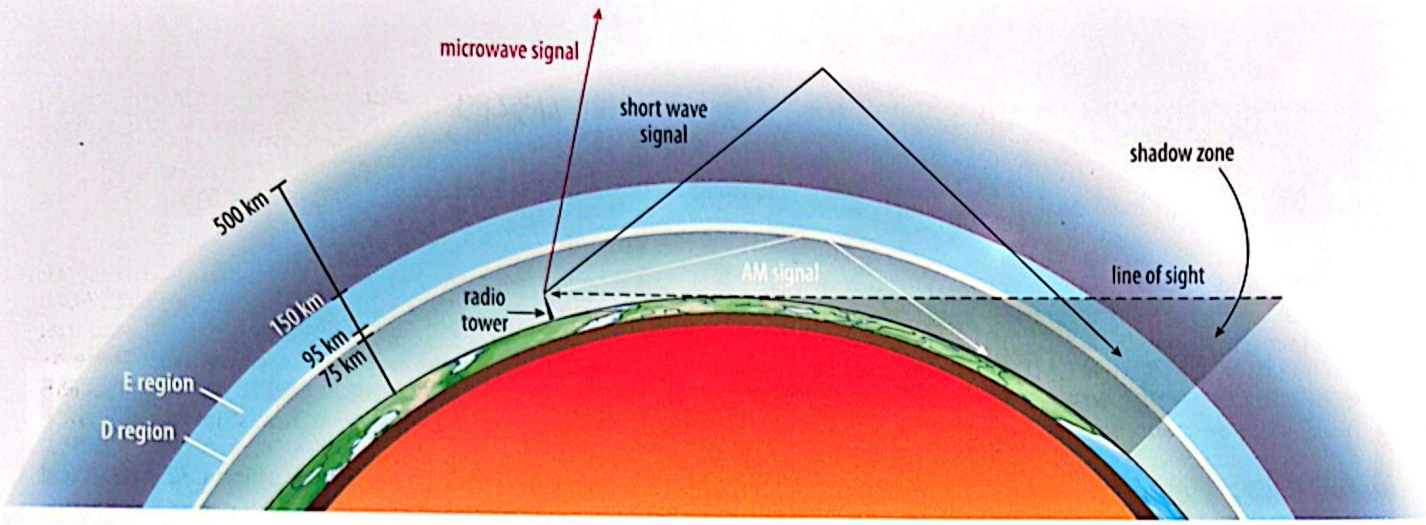
These laboratory findings, seemingly supported by atmospheric observations, propelled the global community into action. The United States, along with several other nations, banned CFC production in 1978 in what was called the *Montreal Protocol*. As satellite evidence of ozone depletion continued to mount, other countries put bans on producing, selling, and buying CFCs in 1987 and 1993. The laws required that they be replaced with other chemicals, but these chemicals didn't work as well as Freon.

So what happened? Is the ozone hole getting smaller? Were scientists right about the link between Freon and the ozone hole? Has banning CFCs really made a difference?

Well, we're not quite sure about any of those questions. Scientists have been watching the ozone hole for several decades now. It has stayed about the same size, but that size is quite variable. In 2003, the hole seemed not to develop at all, perhaps because it was too warm. It may take a long time, maybe another twenty or thirty years or more for CFCs to disappear from the atmosphere. Only then will we be able to tell whether our efforts were actually effective in saving the ozone layer.

It's possible that there are things about the ozone layer that we just don't understand yet. Time will tell whether this was good and wise dominion. Can man's activities really affect the earth and its systems in such a big way? Should we take immediate action to fix what we think are problems with the earth's systems? Do we understand Earth's systems well enough to make decisions about these things without understanding very long-term effects, such as the impact of solar cycles? These are questions that we must answer before we address what many people believe is the coming catastrophe of climate change.

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18-11
The structure and features of the ionosphere

I The ionosphere doesn't reflect microwaves, the waves used in radar and satellite communications. They go right through. This is why NASA uses microwave signals to communicate with spacecraft.

18.7 IONOSPHERE

Above the ozone layer, there's another useful zone in the thermosphere, beginning around 75 km and extending to 500 km above the ground. Here, the sun's rays, as well as rays from space, break some of the atmospheric gas molecules into ions. This layer of ionized atoms forms the **ionosphere**. The ionosphere actually contains up to four layers that combine and change between day and night and with the seasons.

The ionosphere can reflect radio communication waves. Until the second half of the twentieth century, radio users depended almost entirely on this effect to communicate over long distances. Even today, some radio stations and amateur radio broadcasters send their messages around the curve of the earth by bouncing their radio signals off the ionosphere, especially at night. Some shortwave radio signals can return to Earth thousands of miles away from their transmitting antennae. For example, a Christian missionary radio station in Carriacou, an island just off the coast of Venezuela, has listeners in Finland and Russia without using communication satellites!

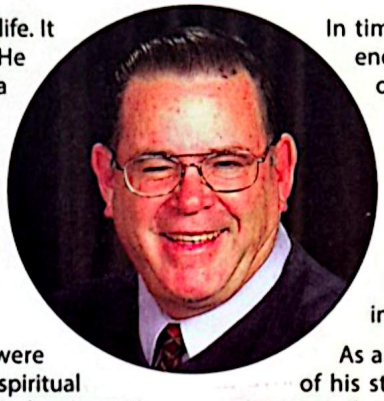
18.8 MAGNETOSPHERE

Yet another amazing zone of the atmosphere is the **magnetosphere** (mag NEE toe SFEER). The magnetosphere extends thousands of kilometers above the earth's surface at distances equal to 10–25 earth radii or more (60,000–160,000 km). This region includes the exosphere of the earth's atmosphere.

James Van Allen discovered the magnetosphere in 1958 using information gathered by America's first artificial satellite, *Explorer 1*. More data revealed several rings or belts where most of the protons and electrons from the sun's solar wind end up trapped in the earth's magnetic field. These regions are fairly close to the earth compared to the outer boundaries of the magnetosphere. In honor of Van Allen's discovery, these regions of dangerous radiation are called the *Van Allen belts*. If you had continued your space elevator trip to the altitude of the Van Allen belts, you would have received a lethal dose of solar

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LARRY VARDIMAN, ATMOSPHERE SCIENTIST



Larry Vardiman was at a turning point in his life. It was 1972, and he was a graduate student. He had joined the US Air Force in 1966 with a bachelor's degree in Physics. The Air Force decided to use him as a weatherman, sending him to do cloud physics research at Scott Air Force Base. Then he moved on to Colorado State University for a PhD in atmospheric science. The decision he had to make in his studies was this—would he remain faithful to the Bible?

The facts and theories that his professors were teaching him contradicted Scripture. After a spiritual struggle, Vardiman decided to hold to his belief in the Bible, even if it went completely against what he learned in the science classroom. Two weeks later, he met Henry Morris.

Henry Morris was one of the founders of the modern Creationist Movement. He and another key creation scientist, Duane Gish, encouraged Vardiman and confirmed the decision he had made. That encounter led him to a career in creation science.

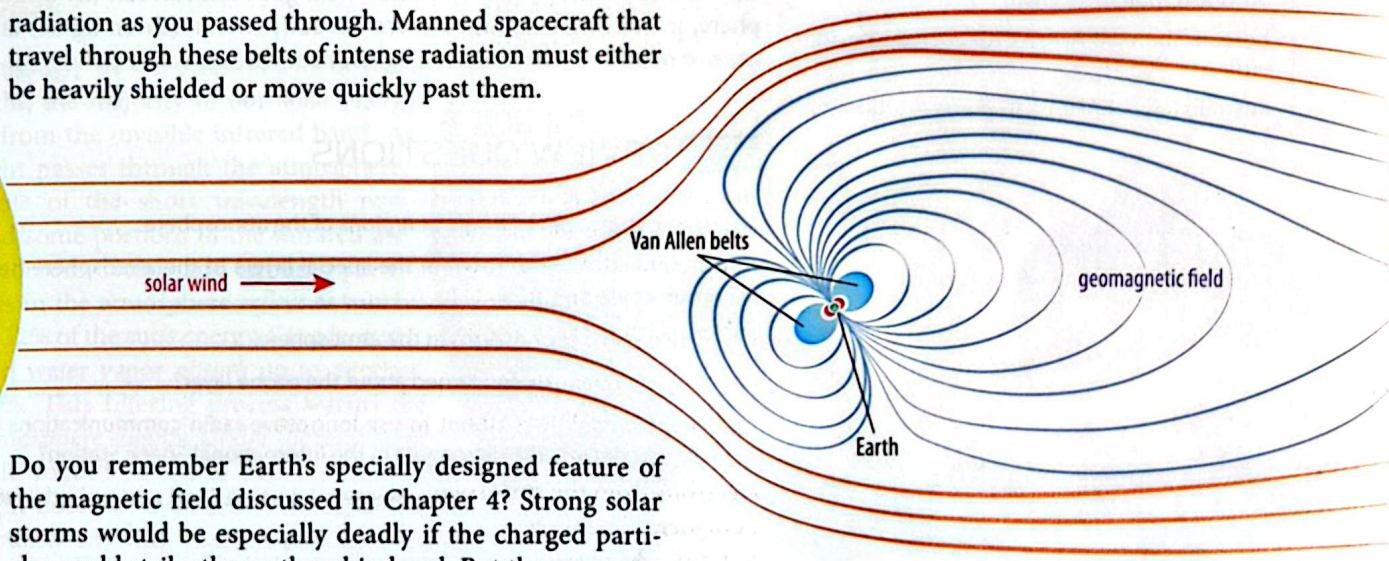
After getting his doctorate degree, Vardiman became the director for the cloud seeding project in northern California. His goal was to bring rain to that drought-stricken area.

In time, Vardiman began to teach math and science at the Institute for Creation Research (ICR), originally near San Diego. ICR is a school that educates many creation scientists. There he began to research, write, and publish scientific articles in creationist journals. Eventually, he became the Chief Operating Officer for ICR. He's on the cutting edge of biblical origins science, especially in meteorology. Most recently, he's targeted the global warming controversy.

As an atmosphere researcher, Vardiman and one of his students tested the idea long held by many Christians that most, if not all, of the water for the Flood came from the collapse of a water vapor canopy. Their computer model of the earth's atmosphere suggested that under a canopy that could have supplied even a small fraction of the water needed for the Flood, living conditions would have been intolerably hot due to a runaway greenhouse effect. These findings show that even biblically based theories can be changed or discarded. Only the Bible is true and unchanging.

Our culture constantly attacks the Bible and its teachings. If you are ever in a situation that makes you question God's Word, do what Larry Vardiman did. The Bible is rock-solid, worthy of your faith and belief!

radiation as you passed through. Manned spacecraft that travel through these belts of intense radiation must either be heavily shielded or move quickly past them.



Do you remember Earth's specially designed feature of the magnetic field discussed in Chapter 4? Strong solar storms would be especially deadly if the charged particles could strike the earth unhindered. But the magnetosphere shields the earth from the brunt of these storms.

18-12
The magnetosphere and the Van Allen belts under normal solar conditions

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18-13
An aurora display over the North Pole

aurora: (L. *aurora*—dawn)

borealis (bore ee AHL is): (Gk. *boreas*, northern)

australis (aw STRAHL is): (L. *auster*—southern)

When the solar wind is especially strong, as it is during a *solar flare* (see Chapter 22), some of the charged particles trapped in the magnetosphere escape near the earth's magnetic poles into the lower regions of the atmosphere. There they collide with gas molecules in the atmosphere, producing beautiful ***aurora borealis*** (the northern lights) and ***aurora australis*** (the southern lights) near the respective poles.

18B REVIEW QUESTIONS

1. Name three special layers or regions of the atmosphere.
2. In general, how do some of the special layers of the atmosphere help protect life on Earth?
3. Explain how ozone forms in the atmosphere.
4. Why are scientists concerned about the ozone layer?
5. Why wouldn't NASA want to use long-wave radio communications to stay in contact with astronauts in the International Space Station?
6. Where do most of the sun's solar wind particles end up trapped in the magnetosphere?
7. State three ways that space travelers can avoid too much dangerous radiation exposure from the Van Allen belts.
8. (True or False) The ionosphere is an important aid to modern microwave satellite communications.

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18C

ENERGY IN THE ATMOSPHERE

How does energy from the sun affect the atmosphere?

18C Objectives

After completing this section, you will be able to

- sketch the flow of energy in the atmosphere.
- compare radiation, conduction, and convection.

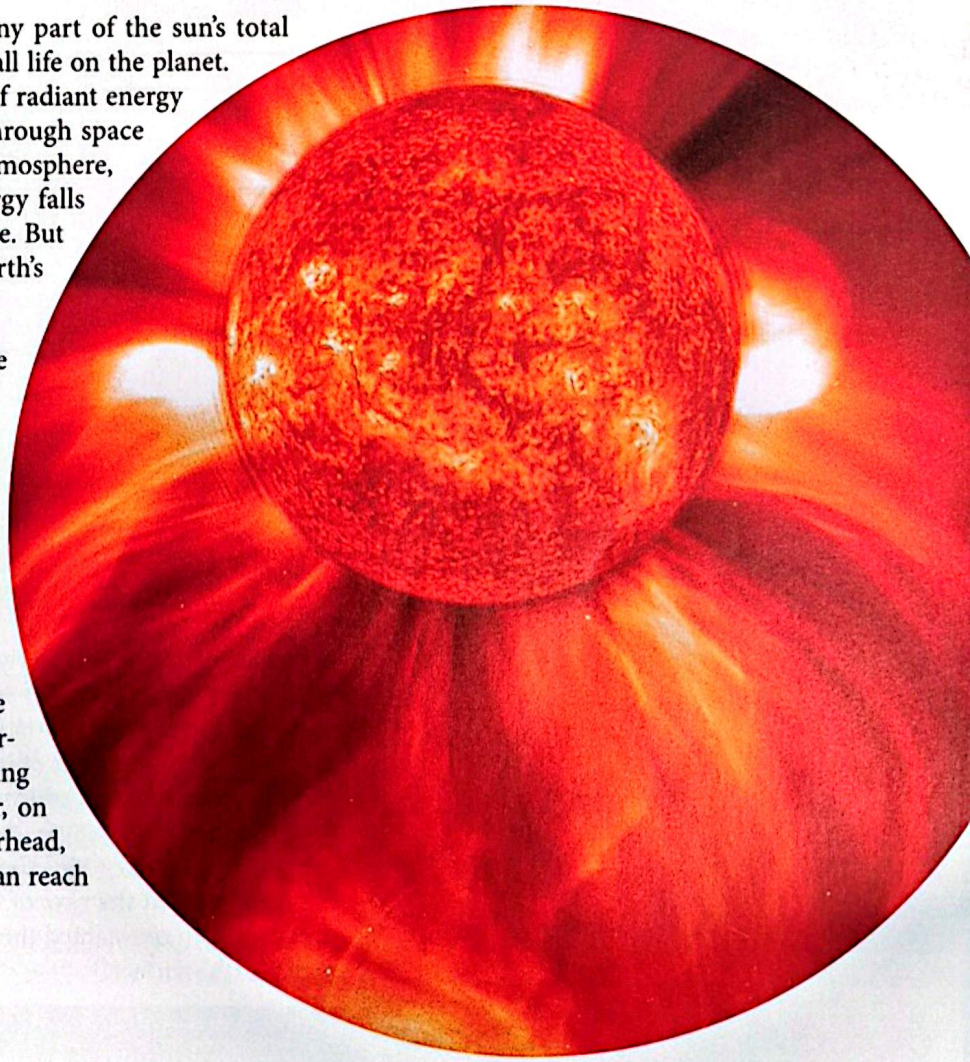
18.9 THE RADIANT SUN

Almost all the energy at the surface of the earth comes directly or indirectly from the sun. The sun emits a wide variety of energy and particles. Most of the sun's radiated energy is in the form of *electromagnetic waves*—light energy. The sun also emits energetic protons and electrons that make up the *solar wind*. In Chapter 2, you learned that light, or radiant energy, comes in different forms (e.g., infrared and visible waves). These different kinds of waves vary in wavelength. Radio waves are the longest waves and gamma rays the shortest. Although the sun emits radiant energy at all wavelengths, most of the radiant energy received by the earth is in the visible and infrared wavelengths.

The sun's energy that reaches the earth's surface on a typical sunny day with the sun directly overhead is about 0.95 kW/m² each second. Following the catastrophic eruption of Mount Tambora in 1815, volcanic dust reduced the surface solar illumination to 80% of that value. This very abnormal condition plunged the world into a "year without a summer."

18.10 THE SOLAR CONSTANT

Although the earth receives only a tiny part of the sun's total energy output, it is enough to sustain all life on the planet. Astronomers call the rate of the flow of radiant energy from the sun that reaches the earth through space the **solar constant**. At the top of the atmosphere, 1.367 kilowatts (kW) of the sun's energy falls on each square meter of the atmosphere. But not all this energy makes it to the earth's surface.



Although the sun gives off a lot of visible light, the majority of our solar energy is from the invisible infrared band. As light passes through the atmosphere, most of the short wavelength rays and some portions of the infrared are absorbed. Clouds and airborne particles in the atmosphere *reflect* as much as 32% of the sun's energy. Clouds, dust, and water vapor *absorb* up to another 18%. This filtering process warms the atmosphere slightly. Under the least favorable conditions, up to half the incoming light does not even reach us! However, on a sunny day with the sun directly overhead, 60–75% of the incoming solar energy can reach the earth's surface.

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18-14
On a normal day, most of the sun's energy that gets to the earth is absorbed by land and water.

When the sun's rays are nearly vertical to the earth's surface, the input of solar energy at the earth's surface is greatest. This can happen only within a narrow band of latitudes between 23½° north and south of the Equator, and then only twice a year at any given location. At other times of the year, and always at other latitudes, the sun's rays hit the ground at some angle, reducing the rate of energy absorbed by the ground, water, or ice. The farther a location is from the Equator, the more slanted the rays of the sun, even at its highest position in the sky. The more slanted the sun's rays, the less energy they deposit at the earth's surface.

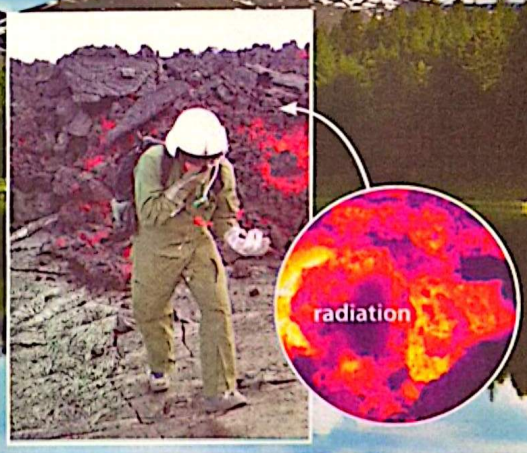
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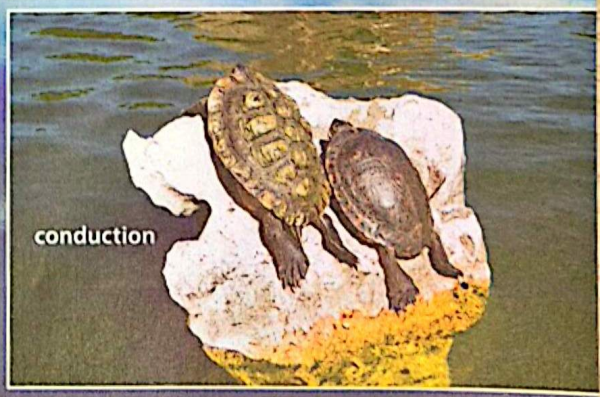
18.11 WARMING THE ATMOSPHERE

As you just learned, the incoming solar energy doesn't heat the atmosphere very much where we need it—in the troposphere. So how does the atmosphere near the earth's surface absorb solar energy? There are three ways. Some of the light energy reflects off the earth's surface and heads back out into space. On its way through the atmosphere, the air absorbs some of this reflected energy. But the earth's surface absorbs most of the original incoming radiant energy and converts it to thermal energy. The warmed surface radiates infrared rays back into the atmosphere. Infrared rays are the same thing as thermal **radiation**. Gas molecules absorb heat from thermal radiation much more easily than they do from light rays.

A second method of heat transfer is direct contact between the surface of the earth and the atmosphere. Gas particles in the air are in constant motion and collide with water, rock, and soil particles, which the sun's radiation has heated. When these collisions occur, kinetic energy transfers to the gas particles. This process raises the temperature of the air in contact with these surfaces. Heat transfer by direct contact is **conduction**. It is the most efficient of the three methods.

The third method of heat transfer involves the vertical movement of warmed masses of air into cooler regions in the atmosphere. This process is **convection**. Convection is a common kind of density current similar to ones that exist in lakes, in the ocean, and within the mantle. Air warmed by radiation or conduction becomes less dense than the surrounding air. Cooler, denser air flows in under the warmer air, lifting it up into the atmosphere, forming a *convection current*. In addition to moving thermal energy to different levels of the atmosphere, convection currents are important in cloud and wind formation. You will learn about these in later chapters.



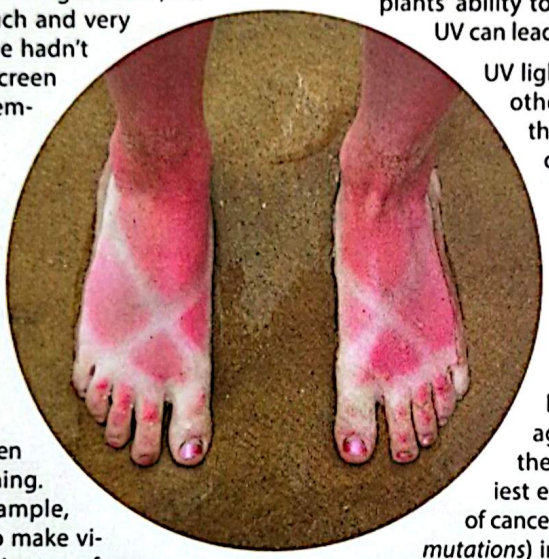




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LIFE CONNECTION: UV LIGHT AND LIFE

Ellen was pumped! It was early June, 80° outside, and the sun was shining brightly—perfect weather for the first day at the beach. She and her friends met for lunch and some time to enjoy the sun and sand. A gentle sea-breeze kept them cool and the hours passed quickly. As she prepared to go home, Ellen noticed that her feet felt sore but thought it was just from the sand. By the time she got home, her feet were bright red, hot to the touch and very tender. She realized her mistake; she hadn't thought that she would need sunscreen this early in the season with the temperatures as low as they were.



Ellen was experiencing one of the negative effects of ultraviolet (UV) radiation. Only a small part of the radiant energy emitted from the sun is visible light—some of it is UV. Ultraviolet energy has wavelengths that are too short for the human eye to see. But we can definitely feel its effects—as sunburn!

Considering Ellen's experience, we often think that UV exposure is a bad thing. But that's not always true. For example, some UV exposure helps our skin to make vitamin D. You need about 10–15 minutes of sunlight a day to build all the vitamin D you need for that day. Vitamin D helps us absorb calcium. Therefore, UV indirectly affects your skeletal development, blood cell formation, immune system function, and many other processes. This is why your mom tells you to go outside and get some fresh air in the sun!

UV is helpful for more than just human nutrition. Studies have shown that it kills germs in the air. UV light also benefits plants. It helps them grow by increasing the amounts of certain hormones. It also helps them fight off pests and bacteria.

However, while small amounts of UV benefit plants, larger amounts will actually stunt their growth or even kill them. This affects not only plants in natural environments, but can also affect farmers' productivity. Because UV can impair plants' ability to perform photosynthesis, too much UV can lead to a smaller crop.

UV light also has bad effects on organisms other than plants. It impairs photosynthesis in some marine organisms. These organisms, called *phytoplankton*, are the basis of the marine food chain, as well as a major source of the oxygen we breathe. So UV can have a snowball effect on ocean organisms right up the food chain.

For humans, the potential problems of overexposure to UV radiation go beyond sunburn. It can prematurely age skin, damage eyes, and suppress the immune system. Probably the scariest effect is its role in causing some types of cancer. Too much UV causes changes (called *mutations*) in a person's DNA, possibly leading to various cancers. Skin cancer is the most common form of cancer in the United States—5.4 million cases every year! A fifth of Americans will develop skin cancer at some point in their lives. Thankfully, you can do a lot to keep from getting skin cancer. The best way is to limit how much time you spend in direct sunlight.

A little ultraviolet light is good for your health. But too much can cause some major health problems. So go catch a few (but not too many) rays!

18.12 THE GREENHOUSE EFFECT

Carbon Dioxide = Air Pollutant?

Many climate scientists and environmentalists think that carbon dioxide is a pollutant because it is a greenhouse gas. They directly link carbon dioxide with global warming and climate change. Not too many of them talk about water vapor this way though. You've probably heard of people talking about how to "reduce your carbon footprint." Living things need carbon from carbon dioxide. We'll discuss this more in Chapter 21 on climate and climate change.

As we have seen, land and water absorb most of the sun's energy that reaches the earth after it passes through the atmosphere. But the earth then indirectly warms the atmosphere through the **greenhouse effect**. The shorter wavelengths of light strike the ground after passing through the clear atmosphere, just like light passing through the glass of a greenhouse. The warmed earth changes the shorter-wave solar energy to longer-wave infrared energy. The infrared energy radiates back toward space, and the lower atmosphere traps and absorbs much of this energy. This is very similar to what happens in a greenhouse. Certain gases that make up the atmosphere, mainly water vapor and carbon dioxide, contribute most to absorbing heat. These kinds of gases are called **greenhouse gases**.

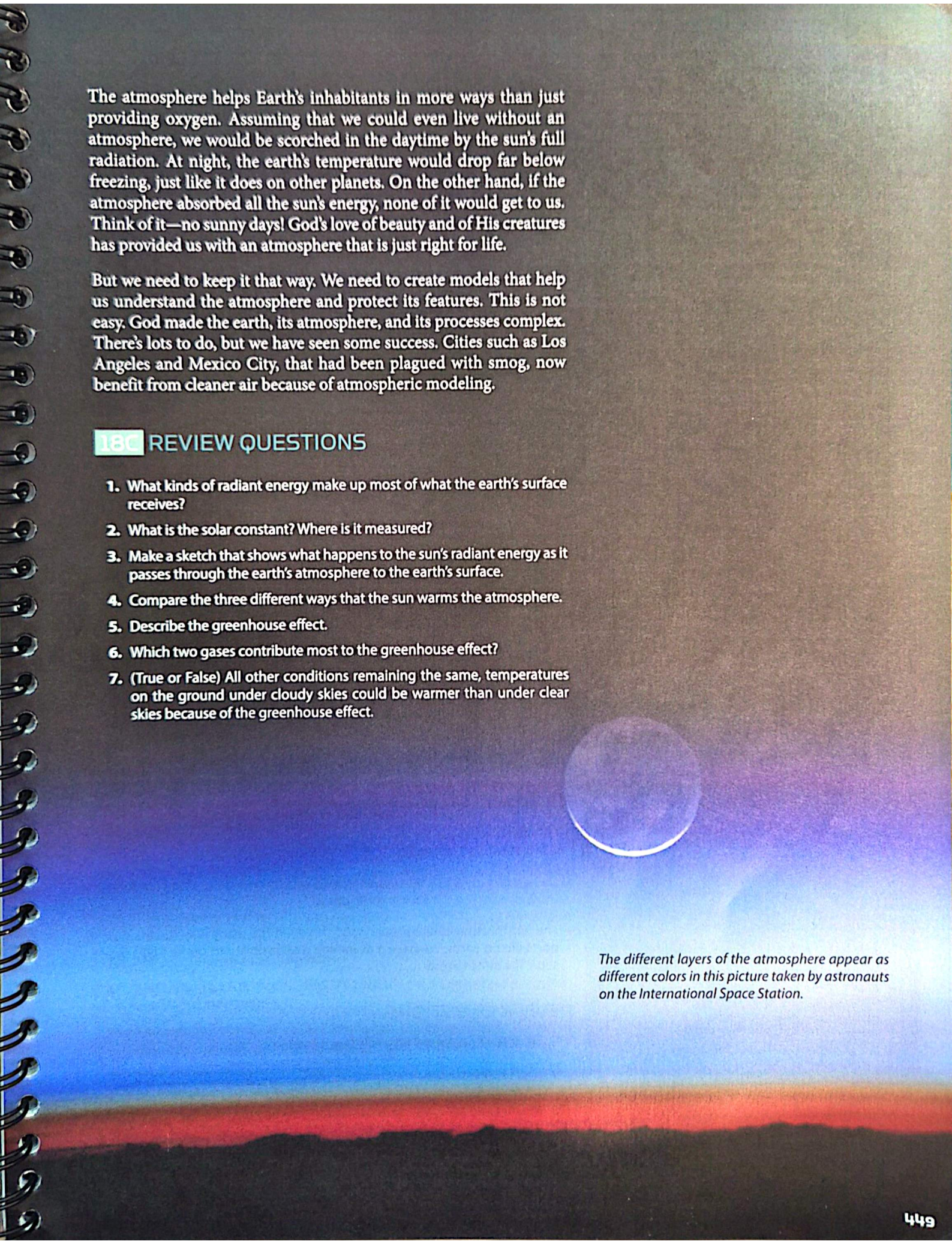
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The atmosphere helps Earth's inhabitants in more ways than just providing oxygen. Assuming that we could even live without an atmosphere, we would be scorched in the daytime by the sun's full radiation. At night, the earth's temperature would drop far below freezing, just like it does on other planets. On the other hand, if the atmosphere absorbed all the sun's energy, none of it would get to us. Think of it—no sunny days! God's love of beauty and of His creatures has provided us with an atmosphere that is just right for life.

But we need to keep it that way. We need to create models that help us understand the atmosphere and protect its features. This is not easy. God made the earth, its atmosphere, and its processes complex. There's lots to do, but we have seen some success. Cities such as Los Angeles and Mexico City, that had been plagued with smog, now benefit from cleaner air because of atmospheric modeling.

18C REVIEW QUESTIONS

1. What kinds of radiant energy make up most of what the earth's surface receives?
2. What is the solar constant? Where is it measured?
3. Make a sketch that shows what happens to the sun's radiant energy as it passes through the earth's atmosphere to the earth's surface.
4. Compare the three different ways that the sun warms the atmosphere.
5. Describe the greenhouse effect.
6. Which two gases contribute most to the greenhouse effect?
7. (True or False) All other conditions remaining the same, temperatures on the ground under cloudy skies could be warmer than under clear skies because of the greenhouse effect.

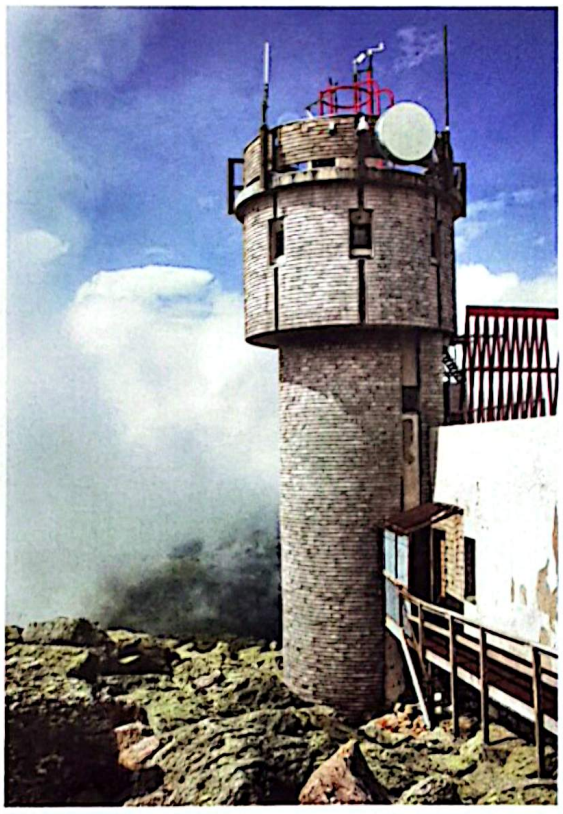


The different layers of the atmosphere appear as different colors in this picture taken by astronauts on the International Space Station.

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Key Terms

atmosphere	429
homosphere	433
heterosphere	433
troposphere	433
lapse rate	433
stratosphere	434
jet stream	434
mesosphere	434
thermosphere	436
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ionosphere	442
magnetosphere	442
solar constant	445
radiation	447
conduction	447
convection	447
greenhouse effect	448
greenhouse gas	448



18-15
When most people think about our atmosphere they think of the weather that occurs there. At an elevation of almost 6300 ft, the Mount Washington Observatory in New Hampshire is often above the clouds.

CHAPTER 18 REVIEW

CHAPTER SUMMARY

» One of the challenges of Earth's growing population is minimizing air pollution in large cities.

» We have a God-given responsibility to wisely care for and use our atmosphere.

» The atmosphere is the envelope of gases that surrounds the earth. Life on Earth would not be possible without it.

» Old-earth scientists believe that Earth's original toxic atmosphere appeared very shortly after the planet formed. After 2 billion years or so, oxygen produced by organisms gradually made the atmosphere able to support more complex forms of life, and biological evolution took off.

» Biblical creationists believe that the earth's atmosphere was probably created no later than Day 2 of Creation week. The atmosphere's ability to protect and sustain life is a compelling case for design by our Creator.

» The two key elements in our atmosphere are oxygen and nitrogen.

» The two regions of the atmosphere differentiated by their composition are the homosphere and heterosphere.

» The regions of the atmosphere that have distinct temperature profiles are the troposphere, stratosphere, mesosphere, thermosphere, and exosphere.

» Nutrients in the earth such as carbon and nitrogen are recycled among land, water, air, and living things.

» The ozone layer and the magnetosphere are special regions lying in the earth's extended atmosphere that absorb or trap harmful solar radiation and particles.

» The ionosphere is a special layer that is useful for certain kinds of global radio communications.

» Not all the sun's energy that reaches the earth makes it to the surface. Nearly half of this energy can be absorbed or reflected by clouds, dust, and atmospheric gases before it reaches the earth's surface.

» The three ways that the atmosphere can absorb solar energy are radiation, conduction, and convection.

» Carbon dioxide and water vapor are two greenhouse gases that contribute most to warming the earth's atmosphere.

» The earth's atmosphere shows good design in the many ways that it supports life on Earth. We have a dominion responsibility to protect and preserve the atmosphere.

REVIEW QUESTIONS

1. What is air pollution? Why is it a problem?
2. What makes Earth's atmosphere different from those surrounding other bodies in our solar system? Why is this significant?
3. In one word for each, state the underlying basis for the two theories of the atmosphere's origin.
4. From where do old-earth scientists think the oxygen in our atmosphere first came?

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5. Create a bar graph that compares the percentage by volume of the gases in the atmosphere.

6. Why are lighter gases higher up in the atmosphere?

7. Use Figure 18-6 to find what temperature layers lie in the heterosphere.

8. What is a lapse rate?

9. What happens to atmospheric temperature above the tropopause in the stratosphere?

10. Defend the following statement: "When we study the earth's atmosphere, we are led to praise God for His care for His creatures."

11. What is the key carbon reservoir in the carbon cycle?

12. Where does organic or fixed nitrogen come from?

13. Using Figure 18-6, identify which temperature layers overlap the ozone layer and the ionosphere.

14. Do any of the special layers of the atmosphere completely block, absorb, or reflect visual light wavelengths? How do we know?

15. Why is the earth's atmosphere not completely transparent to incoming solar radiant energy?

16. In general, how does the solar constant change from the top of the atmosphere to the surface?

17. Make three columns titled *Radiation*, *Conduction*, and *Convection*. Write two everyday examples in each column that are different from those mentioned in the text.

18. How does carbon dioxide contribute to the "greenhouse effect"?

19. What features in the atmosphere does God use to protect His creatures on Earth from extremes of heat and cold?

True or False

20. Christians should be concerned that the world is getting overpopulated with people.

21. Christians should be interested in technology that reduces air pollution.

22. The old-earth model for the origin of the atmosphere is closely related to the nebular hypothesis.

23. All the atmospheric temperature layers with the lowest temperatures have winds and a uniform mixture of gases.

24. The thermosphere is where most of the weather that we experience occurs.

25. The continual mixing of warm and cold air occurs only in the troposphere.

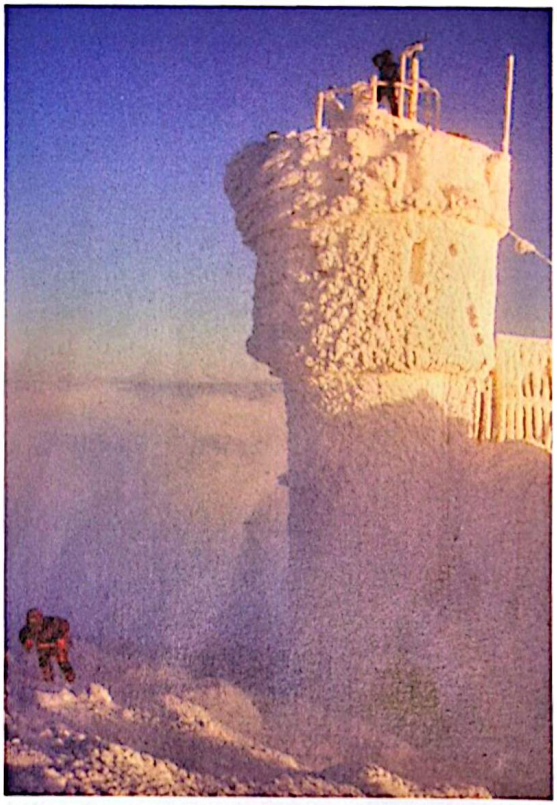
26. The exosphere is mostly hydrogen.

27. The carbon and nitrogen cycles are basically the same but involve different elements.

28. The amount of solar energy reaching the earth's surface on any given day is the same at the Equator as at the Arctic Circle.

29. The sun's energy reaches the earth mainly through conduction.

30. Carbon dioxide is an air pollutant because it is a greenhouse gas.



18-16
Winters can be particularly brutal on Mount Washington, New Hampshire. Rime ice covers the observatory.

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